

Convergence of Random Variables

Probability Theory

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Convergence of random variables is probably the most challenging part of this course.

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1. Real Analysis: the two ideas we need

Before we touch random variables, we need two ideas from real analysis. Everything later is just these two ideas, used on functions that live on a sample space. So it is worth getting them clean first.

1.1 Convergence of a sequence

Suppose we are given a sequence of real numbers $(a_n)_{n \in \mathbb{N}}$, just an infinite list of numbers. We want to say what it means for this list to settle down to a number a .

Definition: convergence of a sequence

$(a_n)_{n \in \mathbb{N}}$ converges to a if

$$\forall \varepsilon > 0, \exists N(\varepsilon) \in \mathbb{N}, \text{ such that } \forall n > N : |a_n - a| < \varepsilon.$$

We then write $\lim_{n \rightarrow \infty} a_n = a$.

In plain words

Pick any small $\varepsilon > 0$. After some point N , every term a_n is within ε of a . The cutoff N can depend on ε : if we ask for a smaller ε , we may have to go further out in the list. That is the whole idea, eventually, and forever after, within ε .

1.2 Pointwise convergence of a sequence of functions

Now we make each term a whole function instead of a number. We are given functions $(f_n)_{n \in \mathbb{N}}$, where each $f_n : I \rightarrow \mathbb{R}$ and $I \subset \mathbb{R}$, and we want to say they converge to a function f . The trick is that we do not compare functions directly. We fix one point and reduce to the case we already understand.

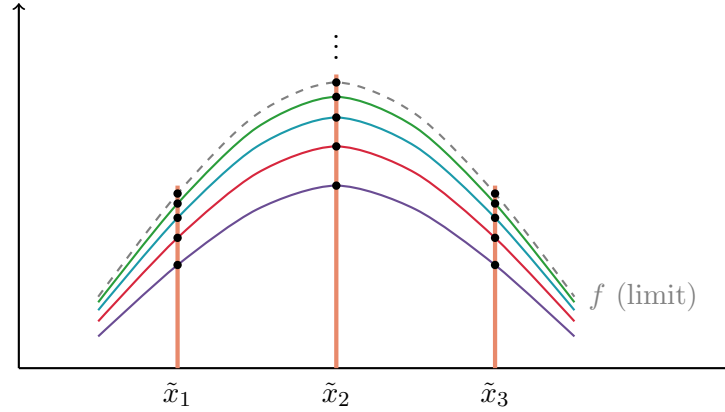
Definition: pointwise convergence

(f_n) converges pointwise to f on I if

$$\forall \tilde{x} \in I, \forall \varepsilon > 0, \exists N(\tilde{x}, \varepsilon) \in \mathbb{N}, \forall n > N : |f_n(\tilde{x}) - f(\tilde{x})| < \varepsilon.$$

In plain words

Fix one point $\tilde{x}$ in I and hold it still. Then the values $f_1(\tilde{x}), f_2(\tilde{x}), f_3(\tilde{x}), \dots$ are just a sequence of numbers, and we ask the question above: does this number sequence converge to $f(\tilde{x})$? If it does at every point $\tilde{x}$, the functions converge pointwise. Note that N depends on both $\tilde{x}$ and ε : we fix the point first, and different points can settle at different speeds.



The curves are the functions $f_1, f_2, f_3, \dots$ (bottom to top), getting closer to the dashed limit f . We show three points $\tilde{x}_1, \tilde{x}_2, \tilde{x}_3$. At each one, the heights where the vertical line cuts the curves form a sequence of numbers $f_1(\tilde{x}), f_2(\tilde{x}), \dots$ (the dots), and they climb up to $f(\tilde{x})$. Pointwise convergence asks this at *every* point of the domain, not only these three.

Key idea

This one move, fix the input and get a sequence of numbers, then use the simple definition, is the idea behind everything below. From now on the outcomes ω will play the role of the points $\tilde{x}$.

2. Back to random variables

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X_1, X_2, X_3, \dots$ be a sequence of random variables. Each one is a function from the sample space to $\mathbb{R}$, that is $X_n : \Omega \rightarrow \mathbb{R}$. So a sequence of random variables is just a sequence of functions, like before.

The only new thing is the probability $\mathbb{P}$. It takes a subset of Ω (an event) and returns a number in $[0, 1]$, the probability of that set. We can think of this number as the size of the set, measured in probability. A rule that assigns sizes to sets like this is called a *measure*, so $\mathbb{P}$ is a probability measure. We will use this in a moment.

Key idea

If we fix one outcome ω in Ω , then every X_n gives a number, and the sequence $X_1(\omega), X_2(\omega), X_3(\omega), \dots$ becomes a sequence of real numbers. Once ω is fixed (picked by nature), there is no randomness left. A fixed ω gives a fixed sequence of numbers, exactly like in real analysis. We call it the sample path at ω .

In general we are not picking ω by sampling. We look at the paths across the outcomes and ask questions about all of them.

3. Sure convergence

Definition: sure convergence

(X_n) converges to X surely if

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \quad \text{for every } \omega \in \Omega.$$

In plain words

This is exactly the pointwise convergence from Section 1, with the sample space Ω as the domain and the outcome ω as the point. Nothing else has changed.

So here is how to read it, slowly.

How to read it

1. Pick any outcome ω and hold it fixed. This turns the random variables into a plain sequence of numbers $X_1(\omega), X_2(\omega), X_3(\omega), \dots$
2. Ask that this number sequence converges to $X(\omega)$, in the ordinary sense: for every $\varepsilon > 0$ there is an N (depending on ω and ε) with $|X_n(\omega) - X(\omega)| < \varepsilon$ for all $n > N$. Past some point, the path at ω stays within ε of $X(\omega)$.
3. Require this for *every* ω in Ω .

Notice that $\mathbb{P}$ does not appear anywhere. Sure convergence is a statement about the functions only. And notice that we are not sampling a random ω and hoping. We are testing every ω , all of them.

Now, why do we call this very strict? Because the limit has to hold at every single ω , with no exception. Even one bad outcome ω , where the path does not settle, breaks sure convergence completely. And in practice there are often such outcomes. Think of tossing a coin forever: one possible outcome is heads every single time, and its path may never settle. Sure convergence fails just because of outcomes like this. This feels wasteful, since that outcome is extremely unlikely. We would like a kind of convergence that survives a few unlikely outcomes. To make that precise we first need to say what unlikely means as a set.

4. What is a set of probability zero?

Recall that $\mathbb{P}$ gives each event its probability, which we are calling its size. A *set of probability zero* (a null set) is a set whose probability is 0. It is so small that the chance of the outcome landing in it is 0.

Example

Take $\Omega = [0, 1]$ with the uniform distribution, so the probability of an interval $[a, b]$ is its length $b - a$. Now look at a single point, say $\{0.3\}$. Its length is 0, so its probability is 0. The chance of landing exactly on 0.3 is 0. The same holds for any single point, and even a countable list of points like $\{0.1, 0.2, 0.3, \dots\}$ still adds up to probability 0.

Key idea

An event of probability 0 effectively never happens. So if something fails only on a set of probability 0, then in every case that has any real chance of happening, it holds. This is the escape we wanted: sure convergence was broken by a few unlikely outcomes, and if those form a set of probability 0, we can allow the limit to fail there and not care.

5. Almost sure convergence

Definition: almost sure convergence (with probability 1)

(X_n) converges to X almost surely if $X_n(\omega)$ converges to $X(\omega)$ on a set of probability 1.

In plain words

“On a set of probability 1” is the whole point. Take the subset of the sample space where the pointwise convergence holds, and call it A . Then “on a set of probability 1” simply means this set has $\mathbb{P}(A) = 1$. We are doing the same test as sure convergence, but instead of asking it everywhere, we only ask that the set where it holds has full probability.

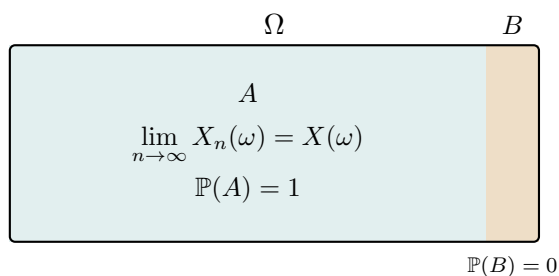
So we split Ω into two parts.

Splitting the sample space

- $A = \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}$ (the limit holds)
- $B = \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\}$ (the limit fails)

These cover everything and do not overlap: $A \cup B = \Omega$ and $A \cap B = \emptyset$.

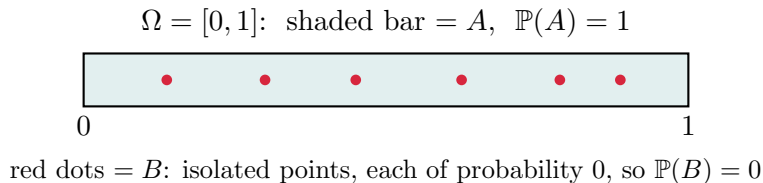
Sure convergence asked $A = \Omega$ (B empty). Almost sure convergence only asks $\mathbb{P}(A) = 1$, which is the same as $\mathbb{P}(B) = 0$. The limit may fail on B , but B must be a negligible set of probability 0.



A first picture, with B drawn as one thin block to keep it simple. The big region A is where the paths converge, $\mathbb{P}(A) = 1$. The thin region B is where they fail, $\mathbb{P}(B) = 0$. Almost sure convergence allows B , as long as it has probability 0.

In that picture B sits in one block. But B does not have to look like that. For a continuous model such as $\Omega = [0, 1]$, the bad set B is usually just a scatter of single points, and each single point has

probability 0, so the whole of B has probability 0 even though it is spread out.



A more honest picture. On $\Omega = [0, 1]$ the bad set B is usually a handful of isolated points (red), not a block. The path converges at every point except these negligible ones.

The notation: build the set, then weigh it

We write almost sure convergence as

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

The limit is *inside* the set. This forces an order of operations, two steps, and you must not mix them.

Step 1. Build the set. Find all ω where the limit holds and put them in A . This step has no probability in it. It is just checking pointwise convergence, ω by ω :

$$A = \{\omega \in \Omega : \forall \varepsilon > 0, \exists N(\omega, \varepsilon) \in \mathbb{N}, \text{ such that } \forall n > N, |X_n(\omega) - X(\omega)| < \varepsilon\}.$$

Step 2. Take the probability. Now, and only now, bring in $\mathbb{P}$ and check $\mathbb{P}(A) = 1$.

So first build the set where the limit holds, then take the probability. Never the other way. If you ever find yourself taking a limit of a probability, you have moved to the next definition.

Note

If the limit holds at every ω (sure convergence), then $A = \Omega$, and $\mathbb{P}(\Omega) = 1$, so $\mathbb{P}(A) = 1$. So sure convergence always gives almost sure convergence. The reverse can fail, because A may leave out a nonempty set B with $\mathbb{P}(B) = 0$.

6. Convergence in probability

Definition: convergence in probability

(X_n) converges to X in probability if

$$\forall \varepsilon > 0 : \lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

Careful: this one is different

In sure and almost sure convergence, the functions (the random variables) themselves converge. In convergence in probability, the functions do not have to converge at all. It is a bit of a

misnomer to say $X_n \rightarrow X$ in probability. What actually converges here is a sequence of numbers (probabilities) going to 0.

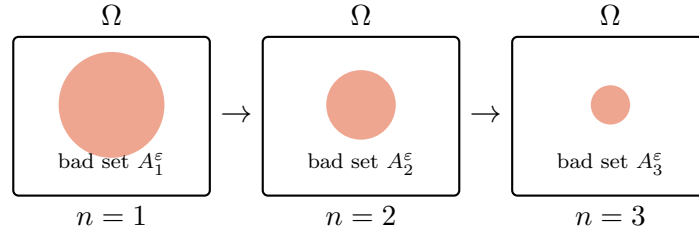
Here is how it works. Fix $\varepsilon > 0$ and hold it. For each n , form the bad set

$$A_n^\varepsilon = \{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\},$$

the outcomes where, at this n , X_n and X differ by more than ε . Take its probability $p_n = \mathbb{P}(A_n^\varepsilon)$, a number in $[0, 1]$. As n grows we get a sequence of numbers $p_1, p_2, p_3, \dots$, and convergence in probability says this sequence goes to 0. So again we are back to the Section 1 definition, a sequence of numbers going to a limit, only now the numbers are probabilities. Compare this with almost sure convergence, where the limit was inside the set and we took the probability once at the end. Here the limit is outside, wrapped around the probability.

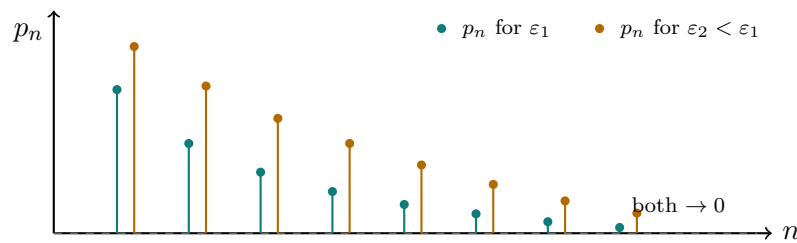
The three steps (for a fixed ε)

1. Look at the event $\{|X_n - X| > \varepsilon\}$. This is the set A_n^ε of outcomes where $X_n(\omega)$ and $X(\omega)$ differ by more than ε .
2. Take its probability $\mathbb{P}(A_n^\varepsilon)$. This measures how big that bad set is.
3. Convergence in probability means this probability goes to 0 as $n \rightarrow \infty$.



At each time n , the bad set A_n^ε is the part of Ω where X_n is more than ε from X . Convergence in probability says its size (probability) shrinks to 0. Here it simply shrinks. In general it can also move around while shrinking, and that moving case is the subtle one in the worked example later.

One more thing about the definition: it must hold for *every* $\varepsilon > 0$. Each ε gives its own sequence p_n , and every one of them must go to 0. For a smaller ε we are stricter about what counts as far, so the bad set is bigger and p_n may be larger, but it still has to go to 0.



For each fixed ε , the probabilities $p_n = \mathbb{P}(|X_n - X| > \varepsilon)$ form a sequence that must go to 0. For a smaller ε_2 (amber) the bad set is bigger, so p_n is larger, but it still goes to 0. The definition asks this at *every* ε at once.

Key idea

One last thing to remember. Almost sure convergence fixes ω and looks at the whole path over all n . Convergence in probability fixes n and looks at the size of the bad set at that n . They are not the same, and we will see an example where one holds and the other fails.

7. The example that separates the two

We now have the two definitions, almost surely and in probability, and on paper they look close. This section gives one sequence where convergence in probability holds but almost sure convergence fails. It is the cleanest example that shows the two are genuinely different, and once you do it by hand the difference is locked in for good.

7.1 The setup

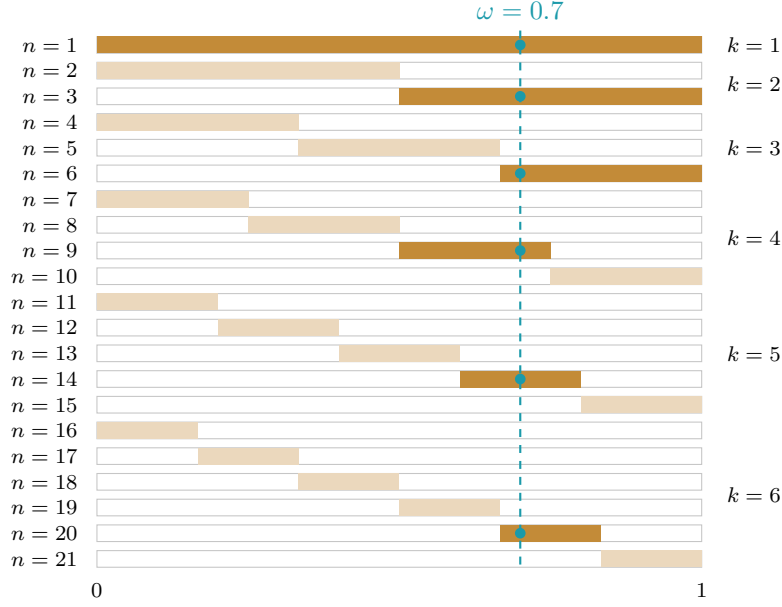
Take $\Omega = [0, 1]$ with probability equal to length, $\mathbb{P}([a, b]) = b - a$. An outcome ω is one point of $[0, 1]$, so picking ω at random is just dropping a pin uniformly on the line. We will build a sequence $X_1, X_2, X_3, \dots$ and ask whether it converges to the zero function $X \equiv 0$.

7.2 The sequence: a sliding bump

We build the intervals block by block. Block k cuts $[0, 1]$ into k equal pieces, listed left to right. Call these intervals $I_1, I_2, I_3, \dots$ in order, and define

$$X_n(\omega) = \begin{cases} 1 & \text{if } \omega \in I_n, \\ 0 & \text{otherwise.} \end{cases}$$

So X_n is 1 on one small interval (the bump) and 0 everywhere else. As n grows, the bump slides across $[0, 1]$, and each time a new block starts it gets thinner. The figure shows the first six blocks, $n = 1$ to 21.



Each row is a copy of $[0, 1]$, and the amber bar is the bump I_n where $X_n = 1$. From top to bottom the bump slides across and gets thinner block by block. The dashed line is one fixed point $\omega = 0.7$. In every block the bump covers 0.7 exactly once (the cyan dots, at $n = 1, 3, 6, 9, 14, 20$). The gaps between those hits keep growing, but the bump always comes back, once in every block, forever.

Writing I_n in terms of n

Which block is n in? The block index k is the one with

$$\frac{k(k-1)}{2} < n \leq \frac{k(k+1)}{2}.$$

This just counts intervals: block 1 holds $0 < n \leq 1$, block 2 holds $1 < n \leq 3$, block 3 holds $3 < n \leq 6$, block 4 holds $6 < n \leq 10$, and so on.

Where inside the block? The position is $j = n - \frac{k(k-1)}{2}$, with $j \in \{1, 2, \dots, k\}$.

The interval and the random variable.

$$I_n = \left[\frac{j-1}{k}, \frac{j}{k} \right], \quad X_n(\omega) = \begin{cases} 1 & \frac{j-1}{k} \leq \omega \leq \frac{j}{k}, \\ 0 & \text{otherwise.} \end{cases}$$

The candidate limit is the zero function, $X(\omega) = 0$ for all ω .

Writing the first ten out, which finishes block 4:

n	1	2	3	4	5	6	7	8	9	10
I_n	$[0, 1]$	$[0, \frac{1}{2}]$	$[\frac{1}{2}, 1]$	$[0, \frac{1}{3}]$	$[\frac{1}{3}, \frac{2}{3}]$	$[\frac{2}{3}, 1]$	$[0, \frac{1}{4}]$	$[\frac{1}{4}, \frac{1}{2}]$	$[\frac{1}{2}, \frac{3}{4}]$	$[\frac{3}{4}, 1]$
$\mathbb{P}(X_n=1)$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

7.3 Convergence in probability holds (the easy side)

This side fixes the step n and measures the size of the bad set. Fix any $\varepsilon \in (0, 1)$. The bad set is where X_n is more than ε away from 0, and since X_n is only ever 0 or 1, being more than ε away

just means $X_n = 1$:

$$\{\omega : |X_n(\omega) - 0| > \varepsilon\} = \{\omega : X_n(\omega) = 1\} = I_n.$$

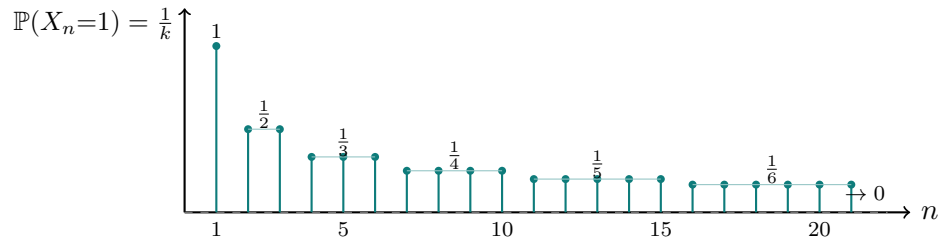
So the bad set is exactly the bump I_n , and its probability is its length,

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(I_n) = \text{length}(I_n) = \frac{1}{k}.$$

Now let n grow. Since $n \approx k^2/2$, the block index k goes to infinity, so $1/k$ goes to 0. Reading the probabilities in order, block after block,

$$\underbrace{1}_{k=1}, \underbrace{\frac{1}{2}, \frac{1}{2}}_{k=2}, \underbrace{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}}_{k=3}, \underbrace{\frac{1}{4}, \dots}_{k=4}, \underbrace{\frac{1}{5}, \dots}_{k=5}, \underbrace{\frac{1}{6}, \dots}_{k=6} \longrightarrow 0.$$

This works for every ε (for $\varepsilon \geq 1$ the bad set is empty, probability 0), so $X_n \rightarrow 0$ in probability. ✓



The bad set is the bump, so its probability is its length $1/k$. Each block contributes a flat run at height $1/k$, and the staircase steps down $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ toward 0. That is convergence in probability: at each step the bad set is small.

Key idea

Convergence in probability only ever looks at one step n at a time. It asks: how big is the bad set right now? Here the bad set is the bump I_n , with size $1/k \rightarrow 0$, so a random pin is very likely on the 0 part. That is all it checks. It never follows a single point through time.

7.4 Almost sure convergence fails (the hard side)

This side does the opposite. It fixes one outcome ω and watches its whole path $X_1(\omega), X_2(\omega), X_3(\omega), \dots$ over all n . Let us do this for a concrete point.

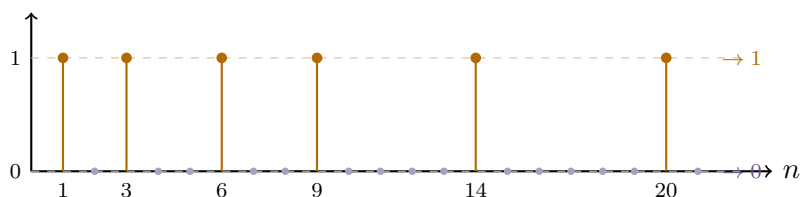
Example

Fix $\omega = 0.7$ and keep it. For each n we just check whether 0.7 is inside I_n :

$$\begin{array}{llll} I_1 = [0, 1] : & 0.7 \in & \Rightarrow & 1 \\ I_2 = [0, \frac{1}{2}] : & 0.7 \notin & \Rightarrow & 0 \\ I_3 = [\frac{1}{2}, 1] : & 0.7 \in & \Rightarrow & 1 \\ I_4 = [0, \frac{1}{3}] : & 0.7 \notin & \Rightarrow & 0 \\ I_5 = [\frac{1}{3}, \frac{2}{3}] : & 0.7 \notin \text{ (} 0.7 > 0.667 \text{)} & \Rightarrow & 0 \\ I_6 = [\frac{2}{3}, 1] : & 0.7 \in & \Rightarrow & 1 \\ \vdots & & & \vdots \end{array}$$

The path begins $1, 0, 1, 0, 0, 1, 0, 0, 1, 0, \dots$, with a 1 at $n = 1, 3, 6, 9, 14, 20, \dots$, that is once in each block.

The pattern is not luck. In block k , the k pieces cover all of $[0, 1]$ and do not overlap, so ω falls into exactly one of them. That means in each block $X_n(\omega) = 1$ happens exactly once. So as the blocks go by, $X_n(\omega) = 1$ happens again and again, once per block, forever.

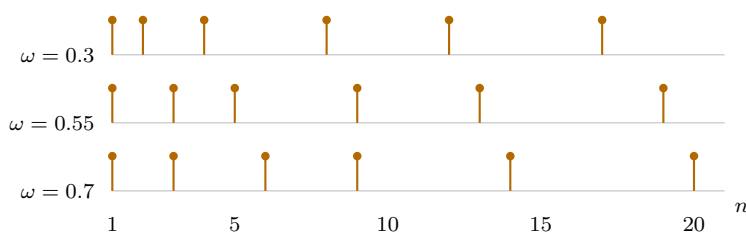


The path $X_n(0.7)$. The amber spikes (value 1) land at $n = 1, 3, 6, 9, 14, 20, \dots$, one per block, getting farther apart but never stopping. Everything in between is 0. The spikes form one subsequence that stays at 1, and the gaps form another subsequence that stays at 0.

Key idea

Why this means no convergence (the subsequence test). Collect the steps where the path is 1 and call them $n_1 < n_2 < n_3 < \dots$. There is exactly one per block, and there are infinitely many blocks, so this list is infinite. Along it $X_{n_j}(\omega) = 1$ for every j , so this subsequence converges to 1. Now collect the steps where the path is 0: another subsequence, and along it the value is 0, so it converges to 0. The 1's appear less often than the 0's, and the gaps between them keep growing, but there are still infinitely many of them. A sequence converges only when every subsequence has the same limit. Here one subsequence goes to 1 and another goes to 0, and $0 \neq 1$, so the path does not converge. In particular it does not converge to 0.

And 0.7 was nothing special. The same thing happens for every ω : in each block the bump covers it exactly once, so its path spikes to 1 once per block and never settles. The next figure shows three different points side by side.



Three different fixed points. Each one is hit by the bump exactly once per block (the amber spikes), so each path spikes to 1 forever and never settles. Whichever ω you pick, you get a pattern like this. So *no* ω has a path that converges to 0.

Key idea

Since this holds for every ω , the good set is empty:

$$A = \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = 0\} = \emptyset, \quad \mathbb{P}(A) = \mathbb{P}(\emptyset) = 0 \neq 1.$$

So X_n does not converge to 0 almost surely. In fact it converges pointwise nowhere, which is even stronger than just failing almost surely.

If you want it as a clean proof, the negation of pointwise convergence does it in three lines. It is enough to find one bad ε , so take $\varepsilon = 1$. Since $X_n(\omega) \in \{0, 1\}$, we have $|X_n(\omega) - 0| \geq 1$ exactly when $X_n(\omega) = 1$. Now fix any ω and any cutoff N . Choose a block k so late that $\frac{k(k-1)}{2} \geq N$, meaning the whole of block k comes after N . In that block ω sits in the piece $j^* = \lfloor k\omega \rfloor + 1$, that is the index $n^* = \frac{k(k-1)}{2} + j^* > N$, and there $X_{n^*}(\omega) = 1$. So no matter how large N is, there is a later n^* with $|X_{n^*}(\omega)| = 1 \geq \varepsilon$. The condition $|X_n(\omega)| < \varepsilon$ for all $n > N$ can never be met. The path does not converge, and this holds for every ω .

7.5 Why this is not a contradiction

The same sequence converges in probability but not almost surely. That feels strange the first time, so here is the reconciliation. Both statements are about the same bumps, but they read them in opposite directions.

- **In probability** fixes the step n and measures the *size* of the bad set: $\text{length}(I_n) = 1/k \rightarrow 0$. At any single step the bump is tiny, so a random pin is almost surely on the 0 part.
- **Almost surely** fixes the point ω and follows its *path* over all n : the bump slides back over ω once per block, forever, so the path never settles.

Key idea

The bump shrinks (its size goes to 0, so in probability holds), but it never stops sweeping back over your fixed point (so almost sure fails). **Shrinking is not the same as leaving.** A small bad set that keeps moving can be tiny at every single instant and still come back to hit every point infinitely often.

Note

In probability (fix n , size of bad set)

bad set = I_n , size = $1/k \rightarrow 0$

holds $\checkmark$

Almost surely (fix ω , whole path)

path spikes once per block, never settles

fails: good set is empty, $\mathbb{P} = 0$

This is the example promised earlier: almost sure is strictly stronger than in probability. Almost sure always gives in probability, but here is a case that has in probability and not almost sure, so the reverse does not hold.